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SMA 2101: Calculus I

LAST REVISION ON October 3, 2013





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 3

Basic Differentiation rules

1) The derivative of a constant

If $f(x) = c$ (a constant) for all x , then $f'(x) = 0$. That is,
$$\frac{dc}{dx} = 0$$

Proof

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{c-c}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{0}{h} \right] = 0$$

2) The power rule

If n is a positive integer and $f(x) = x^n$ then $f'(x) = nx^{n-1}$

Proof

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right]$$

$$f(x) = x^n, \quad f(x+h) = (x+h)^n = x^n + nx^{n-1}h +$$



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$$\begin{aligned}
 & \frac{n(n-1)}{2!}x^{n-2}h^2 + \dots + h^n \\
 \therefore f'(x) &= \lim_{h \rightarrow 0} \left[\frac{(x^n + nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \dots + h^n) - x^n}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[\frac{nx^{n-1}h + \frac{n(n-1)}{2!}x^{n-2}h^2 + \dots + h^n}{h} \right] \\
 &= \lim_{h \rightarrow 0} nx^{n-1} + \frac{n(n-1)}{2!}x^{n-2}h + \dots + h^{n-1} = nx^{n-1}
 \end{aligned}$$

Example $f(x) = 6x^5$

Solution: $f'(x) = 30x^4$ □

EXERCISE 1. $f(x) = x^{10}$

3) The differentiation of a sum

Suppose we know the derivatives of $f(x)$ and $g(x)$

Let $k(x) = f(x) + g(x)$, $k(x+h) = f(x+h) + g(x+h)$



$$k'(x) = \lim_{h \rightarrow 0} \left[\frac{k(x+h) - k(x)}{h} \right]$$

$$\therefore k'(x) = \lim_{h \rightarrow 0} \left[\frac{(f(x+h) + g(x+h)) - (f(x) + g(x))}{h} \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{(f(x+h) - f(x)) + (g(x+h) - g(x))}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{(f(x+h) - f(x))}{h} \right] +$$

$$\lim_{h \rightarrow 0} \left[\frac{(g(x+h) - g(x))}{h} \right]$$

$$f'(x) + g'(x)$$

Example . $k(x) = x^5 - x^4 + x^2 + x + 3$

Solution: $k'(x) = 5x^4 - 4x^3 + 2x + 1$

4) The differentiation of a product

(i) Let $k(x) = \lambda f(x)$



$$k'(x) = \lim_{h \rightarrow 0} \left[\frac{k(x+h) - k(x)}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{\lambda f(x+h) - \lambda f(x)}{h} \right]$$

$$\lambda \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right] \quad \therefore k'(x) = \lambda f'(x)$$

Example . $k(x) = 2x^2 - 3/x + 5\sqrt{x}$

Solution: $k'(x) = 4x + 3/x^2 + 5/2\sqrt{x}$ □

4) The differentiation of a product

(i) Let $k(x) = \lambda f(x)$

$$k'(x) = \lim_{h \rightarrow 0} \left[\frac{k(x+h) - k(x)}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{\lambda f(x+h) - \lambda f(x)}{h} \right]$$

$$\lambda \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right] \quad \therefore k'(x) = \lambda f'(x)$$

Example . $k(x) = 2x^2 - 3/x + 5\sqrt{x}$



Solution: $k'(x) = 4x + 3/x^2 + 5/2\sqrt{x}$ □

(ii) Let $k(x) = f(x)g(x)$, $k(x+h) = f(x+h)g(x+h)$

$$k'(x) = \lim_{h \rightarrow 0} \left[\frac{k(x+h) - k(x)}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{f(x+h)g(x+h) - f(x)g(x)}{h} \right]$$

$$\lim_{h \rightarrow 0} \left[\frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \right]$$

$$\lim_{h \rightarrow 0} \left[\frac{g(x+h)(f(x+h) - f(x)) + f(x)(g(x+h) - g(x))}{h} \right]$$

$$\lim_{h \rightarrow 0} \left[\frac{g(x+h)(f(x+h) - f(x))}{h} \right] + \lim_{h \rightarrow 0} \left[\frac{f(x)(g(x+h) - g(x))}{h} \right]$$

$\therefore k'(x) = g(x)f'(x) + f(x)g'(x)$ (Product Rule)

or $\frac{d}{dx}(uv) = v\frac{du}{dx} + u\frac{dv}{dx}$ (Product rule)

Showing that the differential coefficient of the product of two functions of x is



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equal to the second function multiplied by the differential coefficient of the first

plus the first function multiplied by the differential coefficient of the second.

Remark

If $k(x) = f(x)g(x)h(x)$

then $k'(x) = f(x)g(x)h'(x) + f(x)g'(x)h(x) + f'(x)g(x)h(x)$

5) The differential of a quotient.

Let $k(x) = \frac{f(x)}{g(x)}$

$$k'(x) = \lim_{h \rightarrow 0} \left[\frac{k(x+h) - k(x)}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \right]$$

$$\lim_{h \rightarrow 0} \left[\frac{g(x)f(x+h) - f(x)g(x+h)}{hg(x)g(x+h)} \right]$$





$$\lim_{h \rightarrow 0} \left[\frac{g(x)f(x+h) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x)g(x+h)} \right]$$

$$\lim_{h \rightarrow 0} \left[\frac{g(x)(f(x+h) - f(x)) - f(x)(g(x+h) - g(x))}{hg(x)g(x+h)} \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{g(x)(f(x+h) - f(x))}{hg(x)g(x+h)} \right] - \lim_{h \rightarrow 0} \left[\frac{f(x)(g(x+h) - g(x))}{hg(x)g(x+h)} \right]$$

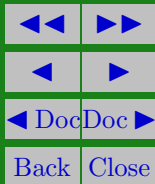
$$= \frac{1}{g(x)g(x)} (g(x) f'(x) - f(x) g'(x)) \text{ (Quotient rule) or}$$

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \quad \text{(Quotient rule)}$$

Thus the differential coefficients of the quotient of two functions of x is equal to the

denominator times the differential coefficient of the numerator times the differential

coefficient of the denominator all divided by the square of the denominator.





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Example . Find $\frac{dy}{dx}$ when $y = \frac{1-x}{1+x}$

$$\begin{aligned}\text{Solution: } \frac{dy}{dx} &= f'(x) = \frac{(1+x) \frac{d(1-x)}{dx} - (1-x) \frac{d(1+x)}{dx}}{(1+x)^2} \\ &= \frac{(1+x)(-1) - (1-x)1}{(1+x)^2} = \frac{-2}{(1+x)^2}\end{aligned}$$

Example . Find $\frac{dy}{dx}$ when $y = \frac{\sin x}{x}$

$$\text{Solution: } \frac{dy}{dx} = f'(x) = \frac{x \frac{d(\sin x)}{dx} - (\sin x) \frac{d(x)}{dx}}{x^2} = \frac{x \cos x - \sin x}{x^2}$$

Example . Differentiate $y = \frac{2}{x} - 5x^2 + \sqrt{x}$

$$\text{Solution: } y = 2x^{-1} - 5x^2 + x^{1/2}$$

$$\frac{dy}{dx} = y' = -2x^{-2} - 10x + \frac{1}{2}x^{-\frac{1}{2}} = \frac{-2}{x^2} - 10x + \frac{1}{2\sqrt{x}}$$

Example . $f(x) = y = (x^2 + \sqrt{x})(\frac{1}{x} - 1)$





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$$\begin{aligned} \text{Solution: } \frac{dy}{dx} = y' = f'(x) &= (x^2 + \sqrt{x}) \left(-\frac{1}{x^2}\right) + \left(2x + \frac{1}{2\sqrt{x}}\right) \left(\frac{1}{x} - 1\right) \\ &= \frac{(-x^2 - \sqrt{x})}{x^2} + \frac{(1-x)}{x} \left(2x + \frac{1}{2\sqrt{x}}\right) = \frac{-(x^2 + \sqrt{x})}{x^2} + 2 - 2x + \frac{1-x}{2x^{3/2}} \end{aligned}$$

□

More Revision Questions

1. Differentiate (Not from first principles)

(i) $y = x^4 + \frac{5}{x^3} + 3\sqrt{x}$

(ii) $y = \frac{x^2-2}{x^3+1}$

(iii) $y = \frac{1-t^2}{t^3}$

(iv) $h(x) = \frac{2x^3+x^2-3x+17}{2x-5}$

(v) $f(t) = \frac{1}{(t+1/t)^2}$

(vi) $g(x) = \left(\frac{1}{x} - \frac{2}{x^2}\right) / \left(\frac{2}{x^3} - \frac{3}{x^4}\right)$

(vii) $f(x) = \frac{x^3 - \frac{1}{x^2+1}}{x^4 + \frac{1}{x^2+1}}$

(viii) $g(y) =$

$2y(3y^2 - 1)(y^2 + 2y + 3)$

(ix) $g(t) = \frac{1}{2t} - \frac{1}{3t^2}$

(x) $f(x) = \frac{2x^3-3x^2+4x-5}{x^2}$

(xi) $f(t) = \frac{4}{(t^2-3)^2}$

(xi) $h(w) = \frac{w+10}{w^2}$



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2. Differentiate from first principles

$$\begin{array}{lll} \text{(i)} & 4x^2 + x & \text{(ii)} \quad \frac{1}{x^2} \\ \text{(iv)} & \frac{1}{2-x} & \text{(v)} \quad \frac{1}{\sqrt{x+2}} \end{array} \quad \text{(iii)} \quad x^3 + x^2$$

The differential coefficients of $\sin x$ and $\cos x$

The functions $f(x) = \sin x$ and $g(x) = \cos x$ are differentiable for all x and that

$$f'(x) = \cos x \text{ and } g'(x) = -\sin x$$

proof:

Recall;

$$1. \quad \lim_{h \rightarrow 0} \sin h = 0 \text{ and } \lim_{h \rightarrow 0} \cos h = 1$$

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = \lim_{h \rightarrow 0} \frac{\cos h}{1} = 1 \text{ (by De L'Hopital's rule)}$$



$$\therefore \lim_{h \rightarrow 0} \frac{\sin h}{h}$$

$$\lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = \lim_{h \rightarrow 0} \sin h = 0$$

$$2. \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \rightarrow 2} x + 2 = 4$$

By De L'Hopital's rule; $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{2x}{1} = 4$

$$3. \lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{x - 2} = \lim_{x \rightarrow 2} x^2 + 2x + 4 = 12$$

By De L'Hopital's rule; $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} \frac{3x^2}{1} = 12$

Recall also the factor

$$1) \cos P + \cos Q = 2 \cos \frac{P+Q}{2} \cos \frac{P-Q}{2}$$



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$$2) \quad \cos P - \cos Q = -2 \sin \frac{P+Q}{2} \sin \frac{P-Q}{2}$$

$$3) \quad \sin P + \sin Q = \sin \frac{P+Q}{2} \cos \frac{P-Q}{2}$$

$$4) \quad \sin P - \sin Q = \cos \frac{P+Q}{2} \sin \frac{P-Q}{2}$$

$$\text{Let } f(x) = \sin x, \quad f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{\sin(x+h) - \sin x}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{\sin x \cos h - \cos x \sin h - \sin x}{h} \right]$$

$$\lim_{h \rightarrow 0} \left[\cos x \frac{\sin h}{h} - \sin x \left(\frac{1 - \cos h}{h} \right) \right],$$

$$\text{NB } \sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} - \sin x \lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = \cos x$$

$$\therefore \frac{d}{dx} (\sin x) = \cos x$$



$$\text{Let } g(x) = \cos x, \quad g'(x) = \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{\cos(x+h) - \cos x}{h} \right],$$

$$\text{NB } \cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$= \lim_{h \rightarrow 0} \left[\frac{\cos x \cos h - \sin x \sin h - \cos x}{h} \right]$$

$$\lim_{h \rightarrow 0} \left[-\sin x \frac{\sin h}{h} - \cos x \left(\frac{\cos h - 1}{h} \right) \right],$$

$$-\sin x \lim_{h \rightarrow 0} \frac{\sin h}{h} - \cos x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = -\sin x (1) - \cos x (0) =$$

$$-\sin x$$

$$\therefore \frac{d}{dx} (\cos x) = -\sin x$$

Example  Differentiate $y = 3x^2 + \cos 2x$ from the first Prin-



principles

Solution: $f(x) = 3x^2 + \cos 2x, \quad f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right]$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \left[\frac{3(x+h)^2 + \cos 2(x+h) - 3x^2 - \cos 2x}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{3x^2 + 6xh + 3h^2 + \cos 2x \cos 2h - \sin 2x \sin 2h - 3x^2 - \cos 2x}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{6xh + 3h^2 - 2 \sin(2x+h) \sin h}{h} \right] = \lim_{h \rightarrow 0} [6x + 3h - 2 \sin(2x + h)] \\ &= 6x - 2 \sin 2x \end{aligned}$$

□

Example . $y = \frac{\cos x}{1 - \sin x}$

Solution: $y' = \frac{-\sin x(1 - \sin x) - \cos x(-\cos x)}{(1 - \sin x)^2}$



$$= \frac{-\sin x + \sin^2 x + \cos^2 x}{(1 - \sin x)^2} = \frac{1 - \sin x}{(1 - \sin x)^2} = \frac{1}{1 - \sin x}$$

□

More Revision Questions

Differentiate

1. $y = \sqrt{x} + \frac{1}{4} \sin(2x)^2$
3. $y = \sin x + \sqrt{\sin x}$
5. $h(x) = 3x^2 - 5 \cos(\pi x^2)$

2. $y = \sin(\cos x)$

4. $h(x) = x^2 + h^{1/x}$



Solutions to Exercises

Exercise 1. $f'(x) = 10x^9$

Exercise 1